

# **Salience-Guided Question Generation: Feature Ablations, Methodological Insights, and Issue Analysis**

---

Madduru Sai Chandra Nikhil

April 2026

# Agenda

- Problem & Motivation
- Proposed Approach & Features
- Experimental Setup & Results
- Qualitative Analysis & Insights
- Limitations & Future Work

# Problem & Motivation

- Standard question generation is often fluent but poorly grounded – questions may ignore key evidence in the passage.
- **SQuAD benchmark:** 100,000+ QA pairs, 3–5 questions per passage, answers are exact extractive spans.
- Challenge: How to steer the LLM toward the most answer-bearing sentences?
- Goal: Improve grounding and answer consistency using sentence-level salience guidance.

*Evaluated on 750 SQuAD passages.*

# Salience-Guided QG Pipeline

1. Score every sentence using linguistic + surprisal features.
2. Select top-k most salient (answer-bearing) sentences.
3. Generate questions from only the selected salient evidence using Qwen2.5-7B-Instruct.
4. Evaluate with overlap metrics + QA consistency.

## Core Design Principle

Soft grounding via salience guidance – prioritize important sentences while keeping questions fluent and diverse.

| Feature Family                                                                | Rationale                                                              |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Linguistic (length, position, POS ratios, lexical density, discourse markers) | Interpretable, fast, strong structural priors for important sentences. |
| GPT-2 surprisal stats                                                         | Captures local uncertainty and information density.                    |
| BERT surprisal stats                                                          | Captures contextual uncertainty with bidirectional context.            |

## Linguistic Features (14 total)

| Category   | Features                                                                          |
|------------|-----------------------------------------------------------------------------------|
| Surface    | sentence_length_words, sentence_position, sentence_position_norm, avg_word_length |
| Lexical    | type_token_ratio, lexical_density                                                 |
| POS ratios | noun_ratio, verb_ratio, adj_ratio, pronoun_ratio, noun_verb_ratio                 |
| Discourse  | named_entity_density, causal_marker_ratio, contrast_marker_ratio                  |

# Experimental Setup (750 Passages)

**Three salience conditions (same prompt, same LLM):**

- **Full Salience:** 26 features (linguistic + surprisal)
- **Linguistic Only:** 14 linguistic features
- **Top-10:** Compact feature set
- **Baseline:** No salience guidance (random sentences)

Same evaluation pipeline for all conditions.

## Salience Classifier: Classification Metrics (2000 Passages)

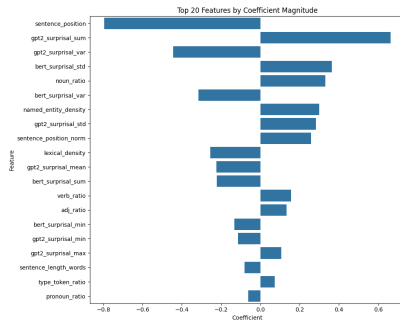
| Configuration            | Features | Accuracy | Precision    | Recall       | F1-Score     |
|--------------------------|----------|----------|--------------|--------------|--------------|
| Full (Ling + Surprisal)  | 26       | 0.693    | <b>0.786</b> | 0.695        | 0.738        |
| No Surprisal (Ling only) | 14       | 0.692    | 0.776        | <b>0.710</b> | <b>0.742</b> |
| Top-10 Compact           | 10       | 0.674    | 0.765        | 0.687        | 0.724        |

**Quick Summary:** Linguistic features are strong; surprisal mainly boosts precision.



# Feature Importance: What Drives Salience?

## Top-ranked features from the 2000-passage salience classifier



**Quick Summary:** The ranking shows that simple linguistic cues remain highly informative, while surprisal features add complementary signal for salience prediction.

# Question Generation Results (750 Passages)

## Salience vs Baseline – 5 Key Metrics

Generated with Qwen2.5-7B-Instruct | Evaluated with deepset/roberta-base-squad2 (QA) & roberta-base (BERTScore)

| Metric             | Salience |       |       | Baseline |              |              |
|--------------------|----------|-------|-------|----------|--------------|--------------|
|                    | Full     | NoS   | Top10 | Full     | NoS          | Top10        |
| ROUGE-L            | 0.349    | 0.360 | 0.363 | 0.350    | 0.369        | <b>0.371</b> |
| QA-EM              | 0.537    | 0.513 | 0.516 | 0.569    | <b>0.578</b> | 0.561        |
| QA-F1              | 0.775    | 0.764 | 0.772 | 0.767    | <b>0.778</b> | 0.765        |
| BERTScore          | 0.905    | 0.908 | 0.908 | 0.906    | 0.909        | <b>0.910</b> |
| Answer-Recovery-F1 | 0.402    | 0.430 | 0.425 | 0.415    | <b>0.454</b> | 0.447        |

**Key Observation:** Baseline wins on surface overlap. Salience wins on answer consistency.

# Salience vs. Baseline Metric Comparison

## Baseline Strength

- Better lexical overlap & BERTScore
- Closer surface match to gold questions

## Salience Strength

- Better QA consistency & answer recovery
- Stronger grounding to important evidence

**Takeaway: Grounding is not the same as gold overlap.**

# Why Questions Get Flagged

## Main reasons (from 2302 issues in qa\_generation\_issues.json)

- Low lexical overlap with gold question
- QA model answer does not match generated answer
- Generated question targets a different (but valid) fact
- QA model answer weakly aligned with gold answers

**Key Insight:** Many flagged questions are not low-quality – they simply focus on different valid facts from the salient sentences.

# Data-Sourced Example 1: Better Semantic Alignment

## Warsaw\_Pact (example\_index 3) — Seeking Specificity

**Gold (SQuAD question):** The UK was initially excluded from participating in which agreement?

### Baseline

Question: Why did the Soviets propose a new agreement?

Answer: to accept the participation of the USA

QA consistency F1: **0.67**

### Saliency (Ours)

Question: What did the Soviets declare about the NATO?

Answer: readiness to examine jointly with other parties

QA consistency F1: **0.94**

*Salient sentence:* “The Soviets declared their readiness to examine jointly...”

**Insight:** Saliency grounds on more specific treaty language, improving QA consistency by 40%. Source: qa\_generation\_issues.json (example\_index 3)

## Data-Sourced Example 2: Grounding Recovers Details

### Warsaw\_Pact (example\_index 10) — Treaty Framework

Gold (SQuAD question): How many countries formed the initial Warsaw Pact membership?

#### Baseline

Question: What did Warsaw Pact members pledge?

Answer: mutual defense

QA consistency F1: **0.36**

#### Saliency (Ours)

Question: What were treaty relations based on?

Answer: mutual non-intervention, sovereignty, independence

QA consistency F1: **0.70**

*Salient sentence:* "Relations among treaty signatories were based on mutual non-intervention..."

**Insight:** Saliency captures nuanced treaty principles; QA consistency improves by 95% via better passage grounding. Source: qa\_generation\_issues.json (example\_index 10)

# Issue-Present Case: Both Methods Flagged

## Pub (example\_index 414)

**Gold (SQuAD question):** In what historical period was a large portion of the population illiterate?

*Salient sentence:* "For this reason there was often no reason to write the establishment's name on the sign..."

### Baseline

Question: Why were pictures useful on signs?

Answer: pictures on a sign were more useful than words

QA consistency F1: **0.0**

### Salience (Ours)

Question: Why was writing the establishment's name on the sign uncommon?

Answer: pictures on a sign were more useful

QA consistency F1: **0.0**

**Takeaway:** This is a true issue case: both methods ask reasonable questions, but they drift from the gold fact target, causing QA mismatch. Source: qa\_generation\_issues.json (example\_index 414)

1. Full features give best consistency; compact features give better overlap.
2. Linguistic features alone are surprisingly powerful.
3. Grounding  $\neq$  gold overlap – salience targets important content.



## Next: Discourse-Level Salience with RST

Current method is sentence-level only. Integrating Rhetorical Structure Theory (RST) will allow us to:

- Identify nuclear (central) vs satellite (supporting) sentences
- Capture discourse relations (evidence, elaboration, contrast)
- Model multi-sentence evidence chains

Expected impact: significantly better salience ranking and consistency on complex questions.

## Conclusion & Open Questions

- Saliency-guided QG is useful when grounding matters.
- Full features improve consistency; compact features improve overlap.
- Next step: optimize for the target objective, not one metric.

Thank you.

Questions, feedback, and suggestions for the next iteration are very welcome.